

What is Subquery and Complex Queries-54

Introduction

In SQL, not all problems are solved with simple SELECT statements. Sometimes you need to build queries inside other queries or combine multiple conditions and aggregations. These are called **subqueries** and **complex queries**.

In MySQL Workbench, subqueries are often used when you cannot directly join tables or when you need intermediate results. Complex queries combine multiple SQL features like subqueries, joins, aggregate functions, and conditions.

Subquery

A **subquery** is a query inside another query.

It is written inside parentheses () and used with SELECT, INSERT, UPDATE, or DELETE.

Types of subqueries:

1. Single-row Subquery

- Returns one row as output.
- Example: Find employees earning more than the average salary.

SELECT Firstname, Salary

FROM salary

WHERE Salary > (SELECT AVG(Salary) FROM salary);

The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying a SQL query with multiple subqueries. The query is as follows:

```
70 • select count(salary) as TotalDept from Salary;
71 • select AVG(salary) as AvgSalary from salary;
72 • select max(salary) as MaxSalary from salary;
73 • select min(salary) as MinSalary from salary;
74 • select Dept, count(*) as Employees, avg(salary) as AvgSalary from salary group by Dept;
75 • select Dept, count(*) as Employees from salary group by Dept having count(*) >= 2;
76 • select Dept, Deptid, count(*) as total from salary group by Dept, Deptid;
77 • select Dept, Avg(salary) As AvgSal from Salary group by Dept order by AvgSal Desc;
```

The 'Result Grid' is visible, showing the results of the query. The columns are 'Dept', 'Employees', and 'AvgSalary'. The data is as follows:

Dept	Employees	AvgSalary
IT-Security	1	21000.0000
Finance	2	24540.0000
Database	2	21500.0000
HR	1	20500.0000
IT-Security	1	19500.0000
Windows	1	27000.0000

The 'Output' tab is also visible, showing the execution log. The log shows the following actions:

- 12 10:34:04 use salary
- 13 10:34:23 select count(*) as Salary from salary LIMIT 0, 1000
- 14 10:36:04 select AVG(salary) as AvgSalary from salary LIMIT 0, 1000
- 15 10:37:11 select max(salary) as MaxSalary from salary LIMIT 0, 1000
- 16 10:37:53 select min(salary) as MinSalary from salary LIMIT 0, 1000
- 17 10:38:50 select Dept, count(*) as Employees, avg(salary) as AvgSalary from salary group by Dept LIMIT 0, 1...

- The subquery (SELECT AVG(Salary)...) returns a single value.

2. Multi-row Subquery

Returns multiple rows. Used with IN, ANY, or ALL.

- Example: Get employees who work in departments located in "Hyderabad".

SELECT Firstname, Deptid

FROM salary

WHERE Deptid IN (SELECT Deptid FROM dept WHERE Location = 'Hyderabad');

For in key word

The screenshot shows the SQL Developer interface with a query window titled 'Query 1'. The query is as follows:

```

77 • select Dept, Avg(salary) As AvgSal from Salary group by Dept order by AvgSal Desc;
78 • select Dept, count(*) as Total from salary where salary > 20000 group by Dept having count(*) > 1;
79 • select count(*) as TotalDept from salary having count(*) >1;
80 • select Dept, Count(*) as TotalDept, AVG(Salary) as AVGSalary from salary group by Dept having Avg(Salary) >20000 order by Total
81 • select Dept, Max(salary) as MaxSalary, Min(Salary) as MinSalary from Salary group by Dept;
82 • select Dept, sum(SALARY) as TotSal from salary group by Dept with rollup;
83 • select Firstname, Salary from salary where salary > (select Avg(salary) from salary);
84 • select Firstname, Dept from salary where Dept in (Select Dept from salary where Deptid = 201);

```

Below the query window, the 'Result Grid' is displayed, showing the results of the last query (line 84). The results are:

Firstname	Dept
Nushina	IT-Security
Sheshadri	IT-Security

At the bottom, the 'Output' window shows the execution log with the following messages:

#	Time	Action	Message
22	10:42:55	select count(*) as TotalDept from salary having count(*) >1 LIMIT 0, 1000	1 row(s) returned
23	10:43:55	select Dept, Count(*) as TotalDept, AVG(Salary) as AVGSalary from salary group by Dept having Av...	3 row(s) returned
24	10:44:47	select Dept, Max(salary) as MaxSalary, Min(Salary) as MinSalary from Salary group by Dept LIMIT 0,...	7 row(s) returned
25	10:45:34	select Dept, sum(SALARY) as TotSal from salary group by Dept with rollup LIMIT 0, 1000	8 row(s) returned
26	10:46:48	select Firstname, Salary from salary where salary > (select Avg(salary) from salary) LIMIT 0, 1000	4 row(s) returned
27	10:47:53	select Firstname, Dept from salary where Dept in (Select Dept from salary where Deptid = 201) LIM...	2 row(s) returned

Subquery returns many Dept values.

For any key word

Query 1

```

78 • select Dept, count(*) as TotalDept, count(*) as TotalDept, AVG(Salary) as AVGSalary from salary group by Dept having Avg(Salary) >22000 order by Total
79 • select Dept, count(*) as TotalDept, count(*) as TotalDept, AVG(Salary) as AVGSalary from salary group by Dept having Avg(Salary) >22000 order by Total
80 • select Dept, count(*) as TotalDept, count(*) as TotalDept, AVG(Salary) as AVGSalary from salary group by Dept having Avg(Salary) >22000 order by Total
81 • select Dept, Max(salary) as MaxSalary, Min(Salary) as MinSalary from Salary group by Dept;
82 • select Dept, sum(SALARY) as TotSal from salary group by Dept with rollup;
83 • select Firstname, Salary from salary where salary > (select Avg(salary) from salary);
84 • select Firstname, Dept from salary where Dept in (Select Dept from salary where Deptid = 201);
85 • select Firstname, Salary from salary where salary >= any (select salary from salary where Deptid = 209);

```

Firstname	Salary
Tanmishte	22500
Athulya	26580
Mitesh	27000
Priya	24500
bharath	24500

salary 20

Output

#	Time	Action	Message
23	10:43:55	select Dept, Count(*) as TotalDept, AVG(Salary) as AVGSalary from salary group by Dept having Av...	3 row(s) returned
24	10:44:47	select Dept, Max(salary) as MaxSalary, Min(Salary) as MinSalary from Salary group by Dept LIMIT 0...	7 row(s) returned
25	10:45:34	select Dept, sum(SALARY) as TotSal from salary group by Dept with rollup LIMIT 0, 1000	8 row(s) returned
26	10:46:48	select Firstname, Salary from salary where salary > (select Avg(salary) from salary) LIMIT 0, 1000	4 row(s) returned
27	10:47:53	select Firstname, Dept from salary where Dept in (Select Dept from salary where Deptid = 201) LIM...	2 row(s) returned
28	10:48:39	select Firstname, Salary from salary where salary >= any (select salary from salary where Deptid = 20...	5 row(s) returned

For all key word

Query 1

```

80 • select Dept, Count(*) as TotalDept, AVG(Salary) as AVGSalary from salary group by Dept having Avg(Salary) >22000 order by Total
81 • select Dept, Max(salary) as MaxSalary, Min(Salary) as MinSalary from Salary group by Dept;
82 • select Dept, sum(SALARY) as TotSal from salary group by Dept with rollup;
83 • select Firstname, Salary from salary where salary > (select Avg(salary) from salary);
84 • select Firstname, Dept from salary where Dept in (Select Dept from salary where Deptid = 201);
85 • select * from Salary;
86 • select Firstname, Salary from salary where salary > all (select salary from salary where Deptid = 209);
87 • select Firstname, Salary from salary where salary > all (select salary from salary where Deptid = 209);

```

Firstname	Salary
Mitesh	27000

salary 21

Output

#	Time	Action	Message
24	10:44:47	select Dept, Max(salary) as MaxSalary, Min(Salary) as MinSalary from Salary group by Dept LIMIT 0...	7 row(s) returned
25	10:45:34	select Dept, sum(SALARY) as TotSal from salary group by Dept with rollup LIMIT 0, 1000	8 row(s) returned
26	10:46:48	select Firstname, Salary from salary where salary > (select Avg(salary) from salary) LIMIT 0, 1000	4 row(s) returned
27	10:47:53	select Firstname, Dept from salary where Dept in (Select Dept from salary where Deptid = 201) LIM...	2 row(s) returned
28	10:48:39	select Firstname, Salary from salary where salary >= any (select salary from salary where Deptid = 20...	5 row(s) returned
29	10:49:50	select Firstname, Salary from salary where salary > all (select salary from salary where Deptid = 209) ...	1 row(s) returned

3. Nested Subquery

A subquery inside another subquery.

- Example: Find employees who earn the maximum salary.

SELECT Firstname, Salary

FROM salary

WHERE Salary = (SELECT MAX(Salary) FROM salary);

The screenshot shows the SQL Enterprise Manager interface. The query window contains the following SQL statements:

```
77 • select Dept, Avg(salary) As AvgSal from Salary group by Dept order by AvgSal Desc;
78 • select Dept, count(*) as Total from salary where salary > 20000 group by Dept having count(*) > 1;
79 • select count(*) as TotalDept from salary having count(*) > 1;
80 • select Dept, Count(*) as TotalDept, AVG(Salary) as AVGSalary from salary group by Dept having Avg(Salary) > 22000 order by Total
81 • select Dept, Max(salary) as MaxSalary, Min(Salary) as MinSalary from Salary group by Dept;
82 • select Dept, sum(SALARY) as TotSal from salary group by Dept with rollup;
83 • select Firstname, Salary from salary where salary > (select Avg(salary) from salary);
84 • select Firstname, Dept from salary where Dept in (Select Dept from salary where Deptid = 201);
```

The results grid shows the following data:

Dept	MaxSalary	MinSalary
IT-Security	21000	21000
Finance	26580	22500
Database	21500	21500
HR	20500	20500
IT-Security	19500	19500
Windows	27000	27000

The output window shows the following messages:

```
# Time Action Message
19 10:40:24 select Dept, Deptid, count(*) as total from salary group by Dept, Deptid LIMIT 0, 1000 8 row(s) returned
20 10:41:11 select Dept, Avg(salary) As AvgSal from Salary group by Dept order by AvgSal Desc LIMIT 0, 1000 7 row(s) returned
21 10:42:05 select Dept, count(*) as Total from salary where salary > 20000 group by Dept having count(*) > 1 L... 3 row(s) returned
22 10:42:55 select count(*) as TotalDept from salary having count(*) > 1 LIMIT 0, 1000 1 row(s) returned
23 10:43:55 select Dept, Count(*) as TotalDept, AVG(Salary) as AVGSalary from salary group by Dept having Av... 3 row(s) returned
24 10:44:47 select Dept, Max(salary) as MaxSalary, Min(Salary) as MinSalary from Salary group by Dept LIMIT 0, 1000 7 row(s) returned
```

4. Correlated Subquery

Subquery depends on the outer query it runs for each row of the outer query.

- Example: Find employees earning more than the average salary of their department.

SELECT Firstname, Salary, Deptid

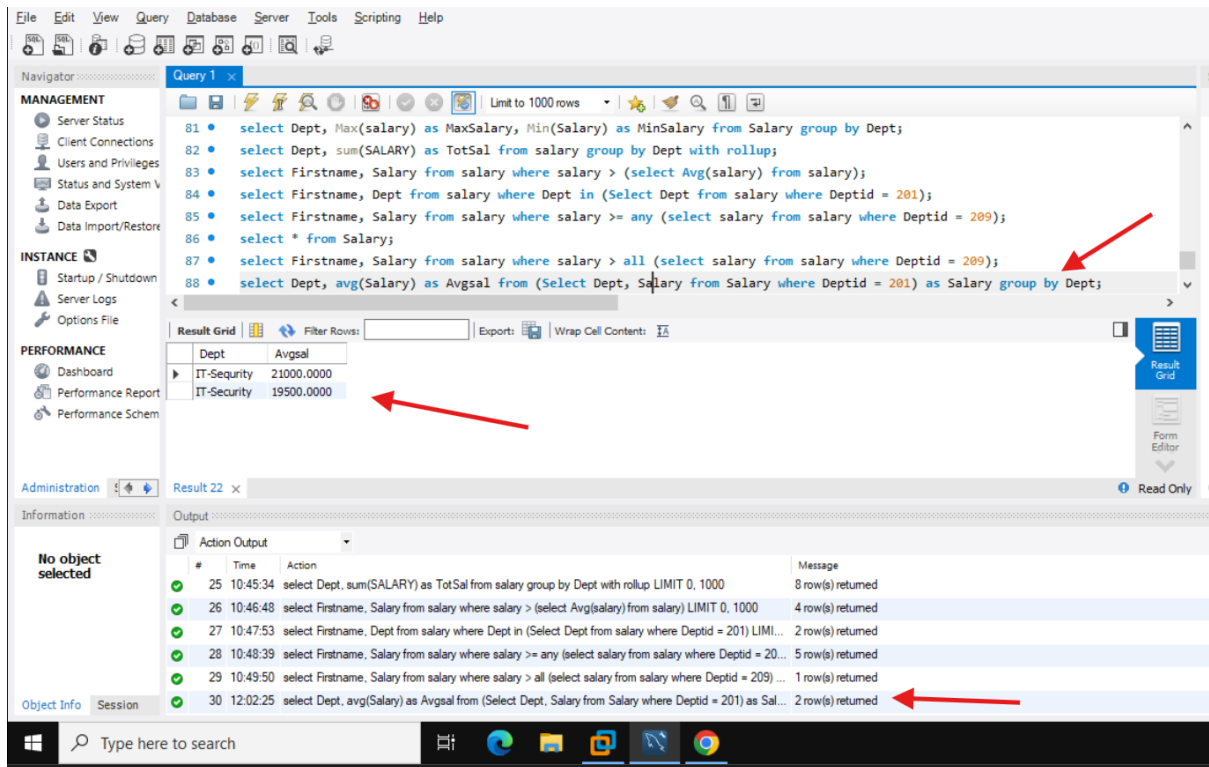
FROM salary s1

WHERE Salary > (SELECT AVG(Salary)

FROM salary s2

WHERE s1.Deptid = s2.Deptid);

- Here, the inner query refers to s1.Deptid.



Complex Queries

A **complex query** is any SQL query that combines multiple concepts: joins, subqueries, grouping, sorting, or conditions.

Examples:

1. Join + Subquery

```

SELECT s.Firstname, s.Salary, d.Dept
FROM salary s
INNER JOIN dept d ON s.Deptid = d.Deptid
WHERE s.Salary > (SELECT AVG(Salary) FROM salary);

```

- Combines JOIN with a subquery filter.

2. Aggregation + Subquery

```

SELECT Deptid, AVG(Salary) AS AvgSalary
FROM salary
GROUP BY Deptid

```

HAVING AVG(Salary) > (SELECT AVG(Salary) FROM salary);

- Uses GROUP BY, HAVING, and subquery together.

3. Nested Queries in FROM Clause

- You can use a subquery as a temporary table.

SELECT Dept, AvgSal

FROM (SELECT Deptid, AVG(Salary) AS AvgSal

FROM salary

GROUP BY Deptid) AS t

JOIN dept d ON t.Deptid = d.Deptid;

- This calculates department-wise average salary and then maps department names.

Key Differences

- **Subquery:** Always a query inside another query.
- **Complex query:** Any query that is not straightforward, may include subqueries, joins, multiple conditions, or aggregate functions.

Why Use Subqueries and Complex Queries?

- Simplify large problems into smaller steps.
- Improve readability by breaking logic into parts.
- Handle advanced reporting like “top N employees by department” or “employees earning above average per location.”
- Allow comparison of data across different sets.